

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

KINETIC THEORY

LECTURE-04 PHYSICS

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Topics To Be Covered

- 1 Revision (dec-1 to dec 3)
- 2 Gabbar Numerical Series (PYQs)
- 3 ..
- 4 ..
- 5 ..

Recep-①

① KTG

$$p = \frac{2}{3} \frac{K \cdot \epsilon}{V}$$

$$K \cdot \epsilon = \frac{3}{2} n R T$$

Mean Free Path



$$p = \frac{1}{3} \rho \underline{v}^2$$

$$p = \frac{1}{3} \rho c^2$$

①

②

③

④

⑤

⑥

⑦

⑧

⑨

⑩

Recap - 2

Gaseous law

$$P \propto \frac{1}{V}$$

$$P \propto T$$

$$V \propto T$$

$$V \propto n$$

Ideal gas equation

$$PV = nRT$$

Mean/Average velocity of gas molecule

$$V_{\text{aim}} = \frac{V_1 + V_2 + V_3 + \dots + V_n}{n} = \sqrt{\frac{8}{\pi} \frac{RT}{m}}$$

Root Mean Square velocity

$$V_{\text{rms}} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + \dots + V_n^2}{n}} = \sqrt{\frac{3RT}{m}}$$

Most probable velocity of gas molecule

$$V_{\text{mpv}} = \sqrt{\frac{2RT}{m}}$$

Recap ③

Degree of freedom

He	O ₂	O ₃
3	5	6

Equipartition of Energy

mole Concept

Application of Specific heat

WORK

Gabbbar

Numerical Series

PYQs

Question-01 [JKBOSE Summer 2026 SET-A]

The mean path of a gas is directly proportional to: *Neet*

①



Pressure



Temperature ✓



The square of the molecular diameter



Number density

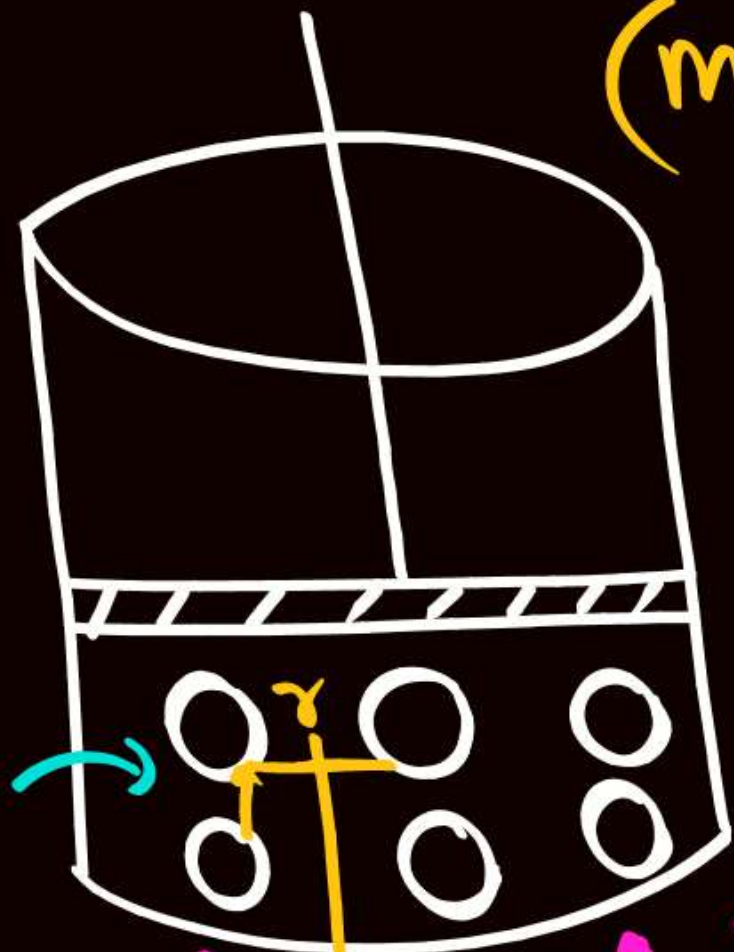


Sol

②

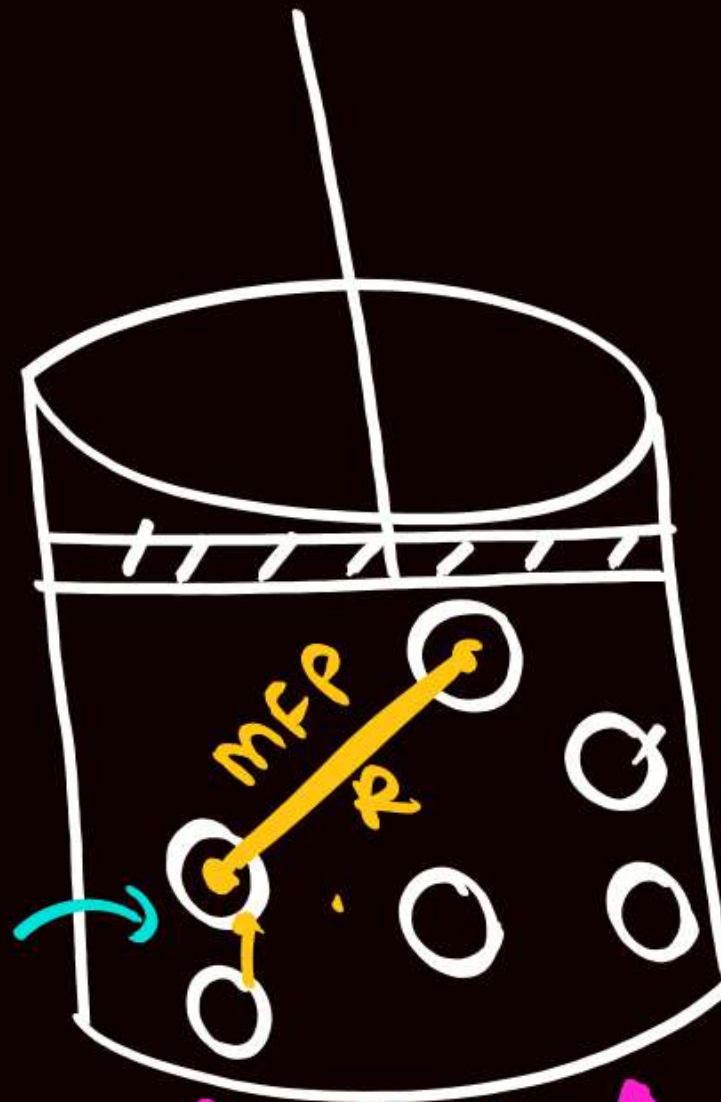
(MFP $\propto$ Temp)

gas



mean free path.

gas



mean free path.

+

Question-02 [JKBOSE Summer 2026 SET-A]

Calculate the root mean square speed of oxygen (O_2) molecules at 273 K.

sol/ $O_2 \rightarrow$ $T = 273 K$ $R = 8.314$ $M = 32 gm$

$T = 273 K$ is in Kelvin

$m = \frac{32}{1000} = 0.032$ kg

$m \rightarrow kg$

$$V_{rms} = \sqrt{\frac{3RT}{m}}$$

$$V_{rms} = \sqrt{\frac{3 \times 8.314 \times 273}{0.032}}$$



$$V_{rms} = \sqrt{\frac{3 \times 8.314 \times 273}{0.032}}$$

$$V_{rms} = \sqrt{\frac{6808.17}{0.032}}$$

$$V_{rms} = \sqrt{\frac{680817}{3.2}} = \sqrt{212755.3} = 461 \text{ m/s} \checkmark$$

Question-03 [JKBOSE Summer 2026 SET-A]

Describe the total degree of freedom for monatomic, diatomic, and triatomic molecules.



Question-04 [JKBOSE Summer 2026 SET-B]

How many particles are present in one mole of any substance?

A

$$6.022 \times 10^{23}$$

$$6.023 \times 10^{23}$$

B

$$6.022 \times 10^{22}$$

C

$$6.022 \times 10^{21}$$

D

$$6.022 \times 10^{24}$$



Question-05 [JKBOSE Summer 2025 SET-A]

What is the value of Avogadro's constant? NQ

Avogadro's constant

$$= \frac{6.022 \times 10^{23}}{\text{mol}}$$

$$= 6.022 \times 10^{23} \text{ mol}^{-1}$$

unit

A

$$6.022 \times 10^{23} \text{ mol}^{-1}$$

B

$$6.022 \times 10^{26} \text{ mol}^{-1}$$

C

$$6.022 \times 10^{25} \text{ mol}^{-1}$$

D

$$6.022 \times 10^{22} \text{ mol}^{-1}$$



Question-06 [JKBOSE Summer 2025 SET-A]

3m ✓



Show that average kinetic energy of translation per molecule of gas is directly proportional to the absolute temperature of gas.

$K.E \propto Temp$



Question-07 [JKBOSE Summer 2025 SET-B]

The S.I. unit of Avogadro's constant is:

$$N_A = \frac{6.022 \times 10^{23}}{\text{mole}}$$

$$N_A = 6.022 \times 10^{23} \text{ mole}^{-1}$$



mol



gram



mol⁻¹



pascal



Question-08 [JKBOSE Summer 2025 SET-C]

How many number of particles are in 22.4 litres of any gas at S.T.P.?

$$\text{no. of mole} = \frac{V}{22.4 \text{ lt}} \quad \text{NO}$$

$$n = \frac{\cancel{22.4}}{\cancel{22.4}} = 1 \text{ mole}$$

$$= 6.022 \times 10^{23}$$

$$= 6.02 \times 10^{23}$$



$$6.02 \times 10^{22}$$



$$2.06 \times 10^{23}$$



$$6.02 \times 10^{23}$$



$$2.06 \times 10^{22}$$



Question-09 [JKBOSE Winter 2025 SET-A]

Write down the four assumptions of Kinetic theory of gasses.

KTG

Assumptions are the points, which are considered without proving it

KTG

1 — 10



Question-10 [JKBOSE Winter 2025 SET-A]

Write down the expression for the pressure of a gas in terms of density.

$$P = \frac{1}{3} \rho c^2$$



Question-11 [JKBOSE Winter 2025 SET-A]

The relation between pressure and energy is:

$K \cdot E$

$$P = \frac{2}{3} \frac{K \cdot E}{V} \rightarrow 1$$

$$P = \frac{2}{3} \frac{K \cdot E}{E}$$

$$P = \frac{2}{3} E$$

A

$$P = \frac{2}{3} E$$

B

$$P = \frac{3}{2} E$$

C

$$E = \frac{1}{2} \rho c^2$$

D

None of them



Question-12 [JKBOSE Winter 2025 SET-A]

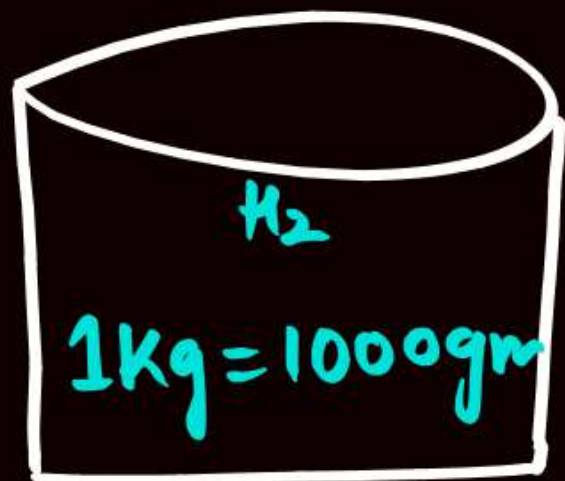
Two vessels A and B of the same size are at the same temperature, one of them holds 1 Kg of H_2 gas and the other hold 1 Kg of N_2 gas. Which of the vessels contains more molecules?

(Neet)

- ☒ A only H_2
- ☐ B only
- ☐ Both (A) and (B)
- ☐ None of them

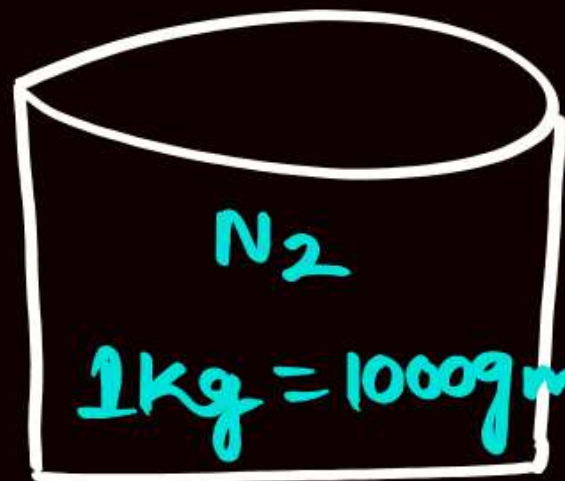


52



$$M_{H_2} = 2\text{gm}$$

Vessel A



$$M_{N_2} = 28$$

Vessel B

$$\uparrow n = \frac{QW}{MM \downarrow} = \frac{1000}{2} = \underline{500}$$

$$\downarrow n = \frac{QW}{MM} = \frac{1000}{28 \uparrow} = \underline{35.7}$$

no. of molecule of H_2 is more than that of no. of molecule of N_2 .

Question-13 [JKBOSE Winter 2025 SET-A]

On the basis of absolute zero as the scale of temperature, the ice melts at:

Kelvin Scale

0°C

$$0 + 273.15$$

$$= 273.15\text{ K}$$

A

100°

B

273.15 K

C

373.15 K

D

None of them



Question-14 [JKBOSE Winter 2025 SET-B]

Two vessels A and B of the same size are at the same temperature, one of them holds 1Kg of H_2 gas and the other holds 1Kg of N_2 gas. Which of the vessels contains more molecules?

A

A only ✓

B

B only

C

Both (A) and (B)

D

None of them



Question-15 [JKBOSE Winter 2025 SET-B]

The expression for the pressure exerted by a gas in terms of density is:



$$\frac{1}{3} \frac{mNc^2}{V^2}$$



$$\frac{1}{2} \frac{mNc^2}{V}$$



$$\frac{1}{3} \rho c^2$$



$$\frac{1}{3} \frac{\rho c^2}{V}$$

$$\frac{1}{3} \rho c^2 \checkmark$$



Question-16 [JKBOSE Winter 2025 SET-B]

Write down the four assumptions of Kinetic theory of gases. ✓



Question-17 [JKBOSE Winter 2025 SET-B]

What is meant by degree of freedom? Write down the degree of freedom of diatomic gas molecule.

3
5
6



Question-18 [JKBOSE Summer 2024 SET-A]

The r.m.s. speed of hydrogen at 27°C is v . What will be the r.m.s. speed of oxygen at 300 K?

(Neet-2019)

v_{rms}



$2v$



$4v$



$v/2$



$v/4$



sol $V_{rms} = \sqrt{\frac{3RT}{M}}$

H₂ $T = 27^\circ C = 27 + 273 = 300K$

$V = \sqrt{\frac{3R \times 300K}{2}}$ — ① $M_{H_2} = 2$

O₂ $T = 300K$ $M_{O_2} = 32$

$V_{rms} = \sqrt{\frac{3R \times 300K}{32}}$ — ②

$\frac{V_{rms}}{V} = \frac{\sqrt{\frac{3R \times 300K}{32}}}{\sqrt{\frac{3R \times 300K}{2}}}$

$\frac{V_{rms}}{V} = \sqrt{\frac{3R \times 300K}{32}} \times \sqrt{\frac{2}{3R \times 300K}}$

$\frac{V_{rms}}{V} = \frac{\cancel{\sqrt{3R \times 300K}}}{\sqrt{32}} \times \frac{\sqrt{2}}{\cancel{\sqrt{3R \times 300K}}}$

$$\frac{V_{rms}}{V} = \frac{\sqrt{2}}{\sqrt{32}}$$

$$\frac{V_{rms}}{V} = \sqrt{\frac{\cancel{2}}{\cancel{32}} \frac{1}{16}}$$

$$\frac{V_{rms}}{V} = \sqrt{\frac{1}{4^2}}$$

$$\frac{V_{rms}}{V} = \frac{\sqrt{1}}{\sqrt{4^2}}$$

$$\frac{V_{rms}}{V} = \frac{1}{4}$$

$$V_{rms} = \frac{V}{4}$$

Question-19 [JKBOSE Summer 2024 SET-A]

Write down four postulates of kinetic theory of gases.

✓ KTG



Question-21 [JKBOSE Summer 2024 SET-C]

The temperature at which r.m.s. of a hydrogen molecule is equal to that of an oxygen molecule at 47°C is:

$$T_{\text{H}} = ?$$

A

80 K

B

-73 K

C

3 K

D

20 K ✓



$$\text{H}_2$$

$$M = 2 \text{ gm}$$

$$\text{O}_2$$

$$M = 32 \text{ gm}$$

$$T = 47^\circ \text{C}$$

$$= 47 + 273 = 320$$

$$V_{\text{rms}} = \sqrt{\frac{3RT_{\text{H}}}{2}}$$

$$V_{\text{rms}} = \sqrt{\frac{3RT_{\text{O}_2}}{32}}$$

$$T_{\text{H}_2} = 2 \times 10$$

$$= 20 \text{ K} \checkmark$$

$$\sqrt{\frac{3RT_{\text{H}_2}}{2}} = \sqrt{\frac{3R \cdot 320}{32}}$$

$$\frac{T_{\text{H}_2}}{2} = \frac{320}{32}$$

7 days challen.....

Unit and Measurement

1D (Basic maths)

2D

HOMEWORK

COPY

H/w



Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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